PHYS 6031
Spring 2024
Homework 3
Due: Friday 9 February 2024 at 4:00 pm in TA's Clippinger mailbox

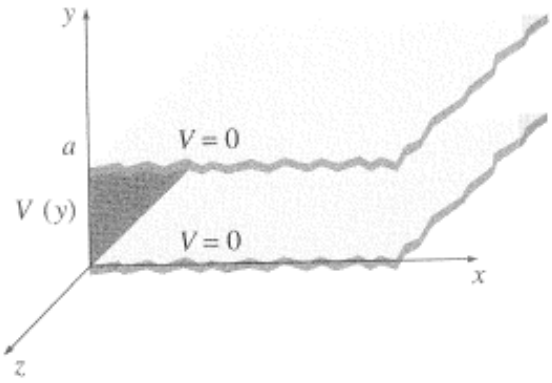
1. Consider the two-dimensional problem where the potential is specified on the the surface of a cylinder of radius b and there is zero charge density inside the cylinder. Show that potential inside the cylinder is given by Poisson's Integral:

$$\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} \Phi(b,\phi')\frac{b^2-\rho^2}{b^2+\rho^2-2b\rho\cos(\phi-\phi')}\,d\phi'.$$

Note that this result implies that the potential at the center of the cylinder is equal to the potential averaged over the surface of the cylinder.

2. Consider a cubical box with side length a. Five of the faces are grounded, while the sixth is held at a constant non-zero potential V_0. Determine the total charge induced on the side *opposite* the side held at a non-zero potential.

3. Two semi-infinite and grounded conducting planes lie parallel to the $x-z$ plane. One is located at $y=0$ and the other at $y=a$. Both planes end at $x=0$ and the gap is closed with an infinite strip maintained a non-zero potential $V(y)$:



One may also assume that $\Phi(x\to\infty, y) = 0$. If

$$V(y) = \begin{cases} +V_0 & y < a/2 \\ -V_0 & y > a/2 \end{cases}$$

and V_0 is a constant, determine

(a) The potential between the planes.
(b) The surface charge density on the lower plane.

4. On the previous homework, you should have found that the stored energy of a uniformly-charged sphere is

$$W = \frac{3}{5}\frac{Q^2}{4\pi\epsilon_0}\frac{1}{R},$$

where Q is the total charge and R is the radius. Let's use this as a model for nuclei, which approximately obey $R = 1.3A^{1/3}$ fm, since nuclear matter has approximately uniform density. Here, A is the number of nucleons (neutrons plus protons). If a Uranium nucleus ($Z = 92$, $A = 235$) splits symmetrically (into two identical nuclei), how much energy (in MeV) is released? Note that symmetric fission is not usually what happens, and and an exactly even split is not even possible in this case since you cannot have half a neutron. Nevertheless, this problem is still a reasonable approximation. Note further: this is the calculation that Lise Meitner performed in late 1938 that provided the essential physical explanation of how fission was possible and indicated fission could provide a very large energy release.